

(a) Project Statement

Wine quality has a significant impact on consumers' willingness to repurchase, long-term brand loyalty, and buying decisions. Product development, quality control, and market positioning all depend on producers knowing what influences perceived quality. The ability to predict consumer reactions to a wine can influence choices about raw materials, processing conditions, and pricing, which directly affects the wine industry's economic growth.

In reality, professional panels or consumers' sensory scores are frequently used to assess wine quality. These scores are laborious and subject to human error, despite being informative. Acidity, residual sugar, sulfur dioxide, and alcohol content are examples of physicochemical measures that can be obtained in a laboratory setting with relative reliability, speed, and objectivity. Producers may be able to monitor and enhance wine quality more effectively by using routine laboratory testing if we can model the relationship between these quantifiable chemical characteristics and sensory quality.

This study specifically seeks to identify the chemical characteristics that are most important in evaluating the quality of white wine, along with the extent to which these data may be utilized to build an effective regression-based prediction model. The primary modeling approach will be multiple linear regression. A full model with all physicochemical predictors is first fitted, followed by an AIC-based reduced model as the second model. Model adequacy will be assessed using residual vs fitted and QQ plots. To assess the stability of the main conclusions, normal and bootstrap confidence intervals will be constructed for 2-3 critical coefficients each.

(b) Data description

The data in this study come from real white Vinho Verde wines produced in northern Portugal. Eleven standard physicochemical characteristics, including numerous acidity measures, residual sugar, sulfur dioxide, and alcohol content, were measured for every wine sample by a qualified laboratory. Simultaneously, the same samples were assessed by a panel of skilled wine tasters who gave overall quality ratings on a 0–10 scale. These paired lab measurements and sensory ratings were assembled by Cortez et al. (2009), who made the dataset available via the UCI Machine Learning Repository.

This dataset gives a solid foundation for examining the relationship between wine chemistry and perceived quality. The large sample size and the diversity in both quality scores and predictors support the linear regression models. These models can evaluate how well the predictors combined explain the quality ratings and estimate the relationship between each chemical feature and quality while maintaining the other factors constant. The data also allows for a meaningful comparison of various model specifications and the application of bootstrap processes to assess the stability and robustness of the main model's conclusions, as all variables are measured on the same wines.

(c) Exploratory data analysis

Each of the 4,898 observations on 12 variables in this dataset corresponds to a single sample of white wine. Eleven physicochemical variables (including fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free and total sulfur dioxide, density, pH, sulphates, and alcohol) and one sensory outcome, which is also the response

variable, quality. The majority of wines are scored between 5 and 6 on the quality scale, which ranges from 3 to 9 with a mean of roughly 5.9. Since all of the variables are numerical, they can be treated as continuous in future modeling without the need for additional dummy variables.

The majority of measurements fall within acceptable physical ranges, according to the basic summary. For instance, alcohol content ranges from 8% to 14.2% with a mean near 10.5%, and residual sugar ranges from 0.6 to 65.8 with a median of 5.2, which suggests a right-skewed distribution and some high-sugar outliers. While pH and sulphates are more tightly concentrated, free and total sulfur dioxide likewise have extended right tails. The dataset has good overall data quality because there are no missing values, and no obviously implausible values are shown by the summary.

Correlations suggest several relationships. Alcohol has the largest positive correlation (around 0.44), so wines with higher alcohol content tend to receive higher quality scores on average. In contrast, density has a moderately negative association with quality (approximately -0.31), indicating that lighter wines are often rated as more popular. The majority of other variables are weakly correlated with quality, but chlorides, volatile acidity, and total sulfur dioxide exhibit weaker negative relationships. Some predictors are strongly related to each other, such as residual sugar with density ($r \approx 0.84$) and free sulfur dioxide with total sulfur dioxide ($r \approx 0.62$), so including both in the same multiple regression model may lead to multicollinearity and unstable coefficient estimates.

(d) Data analysis

The main statistical method in this project is multiple linear regression. Let quality be the expert quality score for wine sample i , and let $x_{i1}, \dots, x_{ip}$ represent the related physicochemical measurements. The basic regression model is

$$quality_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

Keeping all other predictors constant, β_j this model denotes the anticipated change in the mean quality score linked to a one-unit increase in the predictor x_j . The analysis treats the quality score as a numerical response and makes the following assumptions: independent, roughly normal errors, constant error variance, and a nearly linear relationship between the response and the predictors.

Since the quality score is a continuous outcome and there are only 11 primary predictors, multiple linear regression is a suitable model that is not overly complicated. In keeping to explain the relationship between each chemical feature and quality, the linear regression framework also provides conventional inference methods and straightforward interpretations of coefficients. Additionally, residual diagnostics can be used to verify its primary assumptions, and a bootstrap analysis can be used to further evaluate the main conclusions.

To fulfill the study objectives while maintaining the analysis's focus and accessibility, two multiple linear regression models are taken into consideration. The first model is a full linear model, which uses all 11 physicochemical variables as linear predictors of quality. When all other variables are included, this model gives a baseline description of how each measurable attribute is related to quality. The AIC-based reduced model is the second model. Predictors that do not significantly improve model fit are removed using a stepwise process based on AIC, starting with the complete

model. The resulting simplified model has a smaller collection of predictors but strikes a fair balance between explanatory power and parsimony.

The reduced AIC would be considered the main selection criterion, and adjusted R^2 would only be used to confirm that the smaller model does not significantly lose explanatory power compared to the full model.

Only the chosen main model is subjected to model diagnostics. Determining if the conventional linear regression assumptions are reasonably satisfied is the goal. Systematic changes in variability can be found using a residuals vs. fitted values plot. The normality assumption is assessed using a normal QQ plot of the residuals. Combined with the main model findings, these diagnostic checks provide qualitative proof of the main linear regression model's suitability.

A bootstrap approach is used as a supplemental analysis to evaluate the stability of the key coefficient estimations in the main model. Bootstrap samples are produced from the observed dataset by resampling the rows with replacement 1000 times. The main regression model is refitted for every bootstrap sample, and the estimated coefficients for the chosen predictors are recorded. The relevant regression coefficients' 95% confidence intervals are then created using the empirical distributions of these bootstrap estimations. The standard normal confidence intervals from the main model are then compared with the bootstrap intervals.

For the full model, the AIC is 11113.48, and the adjusted R^2 is 0.2803. This model includes all 11 predictors, and the output shows that some predictors that have a large p-value might be removed in the AIC-reduced model.

Next, the `step()` function with AIC was applied, starting from the full model. The final AIC-reduced model removes citric.acid, chlorides, and total.sulfur.dioxide, which all have large p-values as I just mentioned above, and keep the rest of 8 predictors. For this reduced model, the AIC decreases to 11108.29, and the adjusted R^2 is 0.2806, which is slightly higher than the full model.

Overall, the AIC-reduced model uses fewer predictors and gets better AIC and adjusted R^2 values, so I chose the AIC-reduced model as the main model for the rest of the analysis.

After choosing the AIC-reduced model, there are some key coefficients that are related to wine quality. Among all predictors, volatile acidity has the strongest effect. Its coefficient is around -1.89 means higher volatile acidity is linked to lower predicted quality. Alcohol also plays a vital role. Its coefficient is around 0.19, which means that a 1% increase is linked to 0.19 points higher predicted quality, holding the other variables fixed. At this point, white wines with higher alcohol tend to have higher quality ratings. On the other hand, density has a large negative coefficient of around -154 because density is measured on a very fine scale around 1.00. The practical effect is moderate rather than extreme because actual changes in density (for instance, 0.001) only little alter the expected quality (about 0.15 points).

The coefficient for the remaining predictors, like pH, sulphates, residual sugar, and free sulfur dioxide, is relatively small. They are still important, but their practical impact on the predicted quality is not so obvious.

The residuals vs. fitted plot and the normal QQ plot are analyzed to determine whether the main model produces reliable findings. These charts help in determining the satisfaction of the main regression assumptions.

Four distinct diagonal bands could be seen in the residuals vs. fitted plot (Figure 1). The response variable, wine quality, is an integer score, which explains this pattern. Points align in bands because the residual for each score is the score minus the fitted value. There is no obvious strong curve, and the points are distributed pretty uniformly above and below zero. Although there are a few points with significant positive or negative residuals, the residuals' distribution appears to be quite uniform throughout the range of fitted values. Overall, this plot demonstrates that the basic model's linearity and constant-variance assumptions are quite reasonable.

The majority of residuals fall within the straight line, according to the usual QQ plot (Figure 2). A few outliers would be expected under an ideal normal distribution; however, the main portion of the plot closely follows the line, while the points at either end slightly bend away. In contrast to the sample size, these tail deviations are not very significant. In summary, the model diagnostics show no significant violations of the main regression assumptions, and the residuals are close to normal with a few outliers in the tails.

To figure out the consistency of the key coefficients, 95% normal confidence intervals were compared to 95% bootstrap confidence intervals for volatile acidity, alcohol, and density. The normal CI for volatile acidity is around $[-2.10, -1.67]$, and the bootstrap interval is $[-2.12, -1.67]$. For alcohol, the normal CI is $[0.15, 0.24]$, whereas

the bootstrap interval is slightly broader, about [0.08,0.27]. The normal CI for density is [-190, -118], and the bootstrap interval is wider, around [-248, -93].

Thus, the bootstrap intervals are similar to the normal intervals and yield the same findings: the variable is related to the response variable because none of the three key coefficients include zero. This implies that, in terms of sample variation, the main model's results are fairly consistent.

(e) Summary and Discussion

The analysis demonstrates that among the eleven physicochemical variables, alcohol, volatile acidity, and density have the strongest relationships with quality. Even after controlling for the other predictors, higher alcohol levels correlate with higher quality scores, whereas higher volatile acidity and density correlate with lower quality. The other variables have small effects in the reduced AIC model. Residual and QQ plots show no significant violations of linear model assumptions, and the substantial agreement between normal and bootstrap confidence ranges for the important coefficients implies that these conclusions are reasonably reliable.

However, the model does have some limitations. The adjusted R^2 is just around 0.28, leaving much of the variation in quality ratings unexplained. The model's transformations are purely linear, and it does not account for more complex correlations or interactions between chemical variables. Furthermore, the quality score is calculated as a numerical response, despite knowing that it is an ordinal rating, so there is no guarantee that the "distance" between each score is equal.

In future research, one approach would be to include more covariates, such as grape varieties, along with specific sensory descriptions. Moreover, some flexible

approaches, such as generalized additive models, would allow for smooth nonlinear effects of alcohol, acidity, and other predictors, allowing for to identification of optimal ranges or decreases in returns. Tree-based models, such as random forests or boosting, may automatically capture complex interactions and nonlinear patterns, thereby improving prediction accuracy and giving additional measures of variable relevance. Through these methods, we can gain a clearer and more accurate understanding of the quality of the wine.

(f) Reference

Cortez, Paulo, et al. "Wine Quality." *UCI Machine Learning Repository*, 2009, <https://doi.org/10.24432/C56S3T>.

(g) Appendix

Figure 1

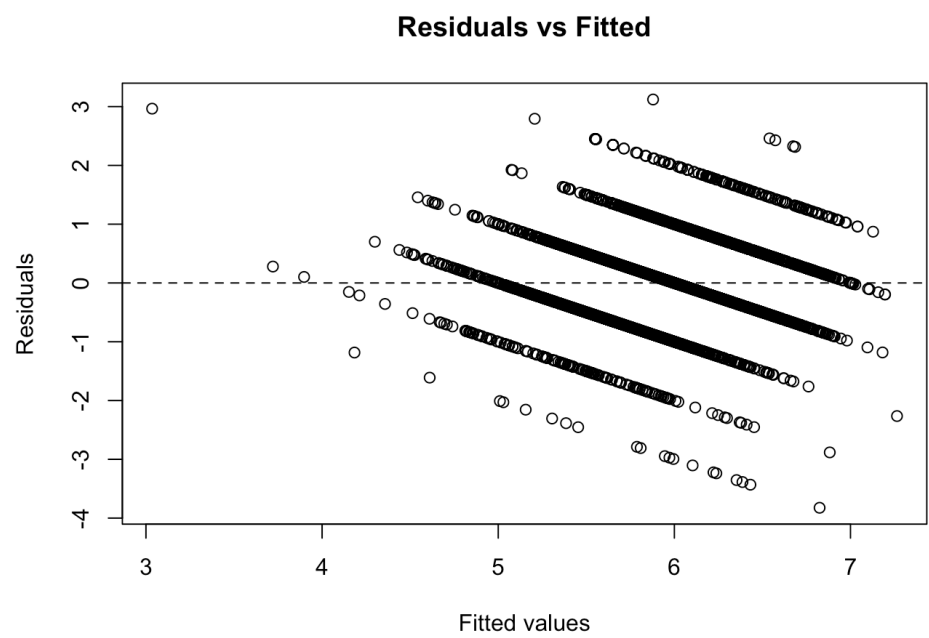


Figure 2

